

Signhify Hunter — Implementation Plan

Codename: HUNTER · **Version:** 1.0 · **Date:** 07 Aug 2026 Build order, milestones, tasks, acceptance criteria. ~7–8 weeks to MVP live, solo-founder-executable with AI pairing tools.

0. Build Strategy

- **Single founder + vibe-coding agents** (Lovable/Cursor/Claude/opencode). Ship vertical slices: every milestone ends with a *working, demoable slice*, not a dead abstraction.
 - Use the God-Level Agent Prompt (`07_GOD_LEVEL_AGENT_PROMPT.md`) as the master build instruction for AI tools; reference these docs for specs.
 - Env gate: email sending is irreversible and reputation-affecting → **everything delivery-related ships last and is staged (sandbox → test domains → live)**.
 - One check per non-trivial unit: Vitest unit tests for verify/scoring/dedup; Playwright smoke for critical flows (login, campaign create, inbox reply).
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1. Milestone Plan (M0–M7)

M0 — Foundations (Days 1–3)

Goal: Repo, infra, CI, compliance skeleton.

- [] Init `signhify/hunter` monorepo: `apps/web` , `apps/api` , `packages/agents` , `packages/db` , `packages/scout` , `packages/outreach` , `packages/shared`
- [] `pnpm` + TypeScript strict + ESLint + Prettier + Vitest + Playwright wired
- [] Supabase project: auth, `app` schema via Prisma migrations, RLS policies, edge function scaffold
- [] Upstash Redis, Resend domain (`hunter.signhify.dev` etc.), Stripe test keys
- [] `.env.example` (all secrets stubbed), CI pipeline (typecheck/lint/test/build)
- [] Compliance pages: unsubscribe endpoint, suppression table, DSAR route, privacy copy
- [] **Accept:** `docker compose up` runs postgres+redis; `pnpm dev` boots API + dashboard skeleton; unsubscribe endpoint returns 200 with test token

M1 — Scout (Days 4–8)

Goal: First real leads flowing from 2 public sources.

- [] `SourceConfig` + `Lead` schema live; dedupe by domain
- [] Adapter A: **HN Algolia API** (who-is-hiring + Show HN) → leads w/ context + source URL

- [] Adapter B: **GitHub API** (search orgs/repos + hiring signals) → leads
- [] Adapter C (stretch): **Reddit JSON** (r/forhire, r/SaaS) with politeness delays
- [] Scout worker: queue → adapter → dedupe → persist → event log; per-source rate limits; retry with backoff
- [] Dashboard: Sources page (config, run status, counts), Leads table (paginated, filter by source)
- [] **Accept:** 2 sources configured via UI; ≥ 100 leads persisted; zero duplicates on re-run; every lead has `source_url`

M2 — Enrich & Verify (Days 9–14)

Goal: 85%+ verified emails.

- [] Enrich worker: company fields (industry, size, tech stack via Wappalyzer-style API or website heuristics)
- [] Verify pipeline: syntax → disposable check → MX → SMTP probe → verdict; catch-all handling
- [] Hunter.io/Apollo API fallback for missing contacts (configurable, budget-capped)
- [] Lead detail view: full enrichment, verdict badge, manual verify retry
- [] **Accept:** batch of 100 leads from M1 → ≥ 85 verified; verdicts visible; re-verify endpoint works

M3 — Qualify (Days 15–18)

Goal: Tier A/B/C with reasons; ICP rules in settings.

- [] Deterministic pre-filters (country, industry blacklist, recency) in SQL
- [] LLM scoring with JSON-schema output; cost-tiered (mini model)
- [] Tier A auto-flags for outreach; Tier B → nurture pool with 14-day re-score job
- [] Leads page: tier filter, score column, expandable reason; manual override
- [] **Accept:** 3 leads manually force-scored via UI in < 5 min; every tiered lead has score + reason + model event logged

M4 — Writer + Campaigns (Days 19–25)

Goal: Compose + run first campaign in sandbox.

- [] Template editor (subject/body with `{{lead.*}}` slots, A/B variants)
- [] Writer agent: per-lead personalization (Claude), 120-word cap, honesty rules, brand voice
- [] Campaigns CRUD + steps (channel, delay, window); audience = saved lead query
- [] Campaign detail: audience size, preview 5 personalized samples, human **Review & Launch**
- [] **Accept:** campaign with 20 Tier-A leads → 20 unique personalized emails generated; preview shows 5; launch gate works

M5 — Delivery Engine (Days 26–32)

Goal: Safe, deliverable sending. (Highest risk — take the time.)

- [] Sending domains mgmt: SPF/DKIM/DMARC setup guide + live DNS check

- [] Warmup engine: 14-day ramp (5→30/day), daily job
- [] Scheduler: timezone-aware windows, global + per-campaign caps, weekday-only default
- [] Resend integration: send, bounces, opens, clicks; bounce handling (hard → suppression)
- [] `list-unsubscribe` + one-click unsubscribe URL in every send; suppression check at send-time
- [] Sandbox mode: send to test inboxes only; then test domains → live at 30/day
- [] **Accept:** campaign sends to sandbox OK; DNS check validates; warmup completes 14 days; manual test to personal inbox lands in Primary with 100% (no spam) on 5 test domains
- [] Deliverability dashboard: SPF/DKIM/DMARC, bounce%, complaint%, blacklist checks

M6 — Respond + Inbox (Days 33–38)

Goal: Replies handled in < 1 hour, HITL gated.

- [] Resend inbound webhook → Thread creation, dedupe by message-id
- [] Auto-stop sequences on reply/unsubscribe; suppression on unsubscribe instantly
- [] Classifier (category + intent + suggestion) with `needs_human` flag
- [] Inbox UI: threads, unread badges (Realtime), reply editor, **Approve & Send**
- [] Positive → Book meeting (Cal.com) + lead status → `meeting` ; CSV export
- [] **Accept:** send test reply to campaign address → thread appears < 30s; sequence halts; approved reply lands back to sender

M7 — Analytics, Billing, Hardening (Days 39–45)

Goal: Ship-ready MVP.

- [] Funnel analytics: sourced → verified → qualified → sent → replied → positive → meeting; per campaign + channel
- [] Usage metering (leads, emails) → Stripe plans (free trial / Pro \$49 / Agency \$149)
- [] Audit log UI (approvals, sends), DSAR processing, retention job
- [] Observability: Sentry + OTel → Axiom; health endpoint; DLQ replay UI
- [] Playwright E2E suite green; seed script with fixture leads
- [] **Accept:** full funnel renders; billing test checkout works; zero PII in logs; all E2E green
- [] **GO LIVE:** switch warm domains to production volume (≤ 150 /day/domain), flip `live` flag

2. Week-by-Week Summary

Week	Milestone	Deliverable (demoable)
1	M0+M1	Repo + dashboard skeleton + 100 leads from 2 sources
2	M2	Enriched, verified lead base
3	M3+M4	Tiered leads + campaign composer with AI personalization
4	M5 (part)	Warmup running, sandbox sends green

Week	Milestone	Deliverable (demoable)
5	M5 (part)	Live delivery at 30/day, deliverability dashboard
6	M6	Inbox with HITL replies
7	M7	Analytics + billing + hardening
8	Go-live ramp	Scale to 150/day, first booked meetings

3. Build Order Rules (for AI coding agents)

1. **Never build UI before its API exists** — implement API route + types first, then page.
2. **Prisma schema is the contract** — update schema → migrate → regenerate types → then code.
3. **Agents are pure functions over queues** — no agent imports a dashboard component; shared package only.
4. **Delivery code is behind a `SANDBOX` env flag** until M5 acceptance.
5. **Every queue handler is idempotent** (natural key upsert) — replays must be safe.
6. **Compliance code first-class**: unsubscribe + suppression exist in M0 and are never bypassed.

4. Testing Strategy

Layer	Tool	Coverage
Unit	Vitest	verify pipeline, dedupe, scoring pre-filters, writer template engine, scheduler math, suppression
Agent	Vitest + mock LLM	JSON-schema parse failures, fallback tier, token caps
API	MSW + Vitest	All <code>/api/*</code> routes, RLS behavior via service-role test user
E2E	Playwright	login, source config, campaign launch (sandbox), inbox reply
Deliverability	Manual staged	warmup completion, inbox placement on 5 test domains, unsubscribe latency
Security	Manual + tools	RLS attempt with 2 users, secret scanning in CI, no PII in logs

5. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Deliverability collapse	Med	High	Warmup $\geq 14d$, $\leq 30/day$ ramp, complaint alarms at 0.1%, domain rotation

Risk	Likelihood	Impact	Mitigation
LLM cost blowup	Med	Med	mini-model bulk, per-campaign budget cap + alarm
Source ToS/blocking	High	Med	Adapters use public APIs/feeds; graceful 429 handling; diversify channels
Platform complaint / legal	Low	High	Compliance built-in (M0), honest personalization, suppression permanent
Solo-founder overload	High	Med	AI pairing for build; HITL gates keep founder time $\leq$ 2h/wk
Data privacy breach	Low	High	RLS, no PII in logs, retention 90d, DSAR endpoint

6. Definition of Done (MVP)

1. Continuous sourcing $\geq$ 2,000 leads/mo from $\geq$ 4 channels.
2. $\geq$ 85% verified emails; Tier-A $\geq$ 8% of sourced.
3. Campaigns run unattended within limits; sequences auto-stop on reply.
4. Inbox replies triaged $<$ 1 min; founder approves sends; bookings created in 2 clicks.
5. Deliverability: $>$ 97% inbox, $<$ 0.1% complaints — measured on dashboard.
6. Full compliance surface live (unsubscribe, suppression, DSAR, retention).
7. Billing live; first 3 paying agency customers (Phase 7 starts, schema already multi-tenant).